

Programação Paralela

Assignment 1

Departamento de Informática
Universidade de Évora

2025/2026

The goal of this assignment is to study the performance of a parallel sort algorithm, implemented for a shared-memory system.

1. Implement a parallel sorting algorithm, or a parallel version of a sequential sorting algorithm, for a shared memory system.

2. Build a program that reads a sequence of signed integer values from its standard input, one per line, sorts the sequence read in non-decreasing order using the algorithm you implemented, and outputs the sorted sequence on its standard output, one value per line.

You may consider that the first line of the input contains the length of the sequence.

3. Alternatively, the program may read the sequence from a file whose name is given on the command line (in which case, the file only contains the sequence of values to be sorted, one per line) and write the sorted sequence into a file whose name is derived from the name of the input file.

4. The program should also be able to receive, on the command line, the number of threads to use to run the algorithm, using the syntax ‘`-t number-of-threads`’:

```
./my-sort -t 16
```

5. The assignment may be completed individually or by groups of two students. In the latter case, two different algorithms must be implemented.

6. To complete the assignment, you must submit the code of your program(s) and a report containing:

- A high-level description of the algorithm(s) implemented.
- The description of the strategy behind the parallelisation of the algorithm(s).
- The analysis of the performance of the algorithm(s) implementation(s), taking into account the time taken by the different phases of the program(s), and including the study of the strong and weak scalabilities of the program(s).
- The comparison of the observed parallel performance with the expected behaviour of the algorithm(s).
- The comparison of the behaviours of the algorithms, in the case of two-person groups.
- Recipes to compile and run your program(s).

7. The preferred programming languages to use for the assignment are C and Java. If you would like to use another programming language, please check with me (vp@...) if it is alright.

For two-person groups, both algorithms should be implemented in the same programming language, so it is easier to compare them.

8. The assignment should be submitted, via moodle, by **May 15, 2026**.